

Learning Discriminative Representations to Interpret Image Recognition Models

Thèse de Doctorat

Felipe Torres Figueroa

École Centrale de Marseille

Laboratoire d'Informatique et de Systèmes (LIS)

Marseille, September 23rd 2024

Jury Members

President: Frédéric Jurie

Examiners: Frédéric Jurie
Giorgos Toliás
Frédéric Precioso
Diane Larlus

Supervisor 1: Ronan Sicro
Supervisor 2: Stephane Ayache
Supervisor 3: Yannis Avrithis

Table of Contents

- Introduction
- 1 Background
- 2 Opti-CAM: Optimizing saliency maps for interpretability
- 3 CA-Stream: Attention-based pooling for interpretable image recognition
- 4 A learning paradigm for interpretable gradients
- References

Table of Contents

- Introduction
- 1 Background
- 2 Opti-CAM: Optimizing saliency maps for interpretability
- 3 CA-Stream: Attention-based pooling for interpretable image recognition
- 4 A learning paradigm for interpretable gradients
- References

Motivation

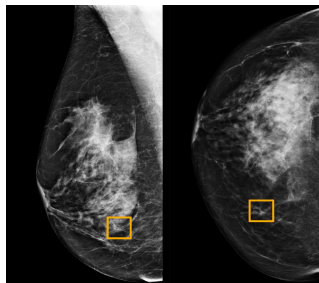
High stakes

Consider these situations:

Motivation

High stakes

Consider these situations:



Motivation

High stakes

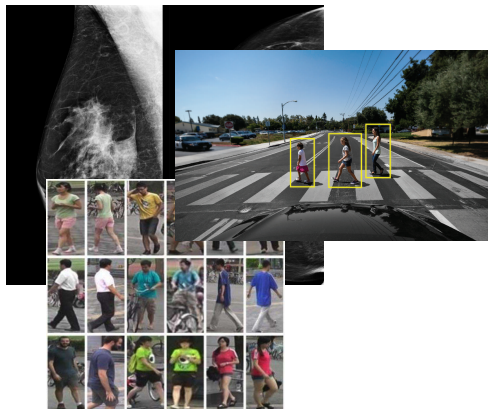
Consider these situations:



Motivation

High stakes

Consider these situations:



Straight to the point



- How do we **know how** a system works?
- How do we **know how** safe a system is?
- If a system fails, **who** is accountable?

We must **understand** the behaviour of these models.

Step by step

Computation, Computer Vision and AI



Explainable AI



Thesis objectives

Computation



Better known as *Computer Science*.

Study of:

- Algorithms.
- Data structures.
- Design of hardware and software.

Computation, Computer Vision and AI

Computer Vision

Replication of human vision capabilities.

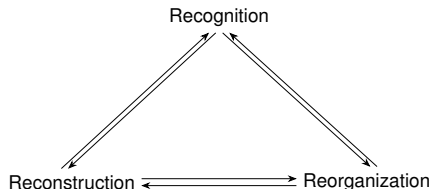
Computation, Computer Vision and AI

Computer Vision

Replication of human vision capabilities.

Three fundamental tasks[1]:

- Recognition.
- Reconstruction.
- Reorganization.



Computation, Computer Vision and AI

Artificial Intelligence

Systems capable of performing tasks requiring human intelligence [2].

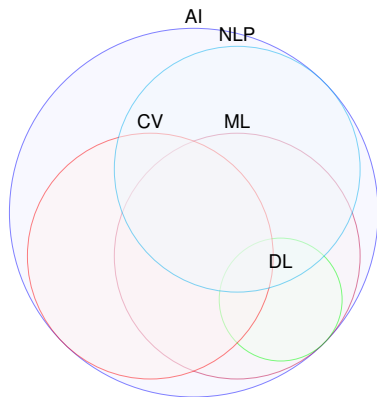
Computation, Computer Vision and AI

Artificial Intelligence

Systems capable of performing tasks requiring human intelligence [2].

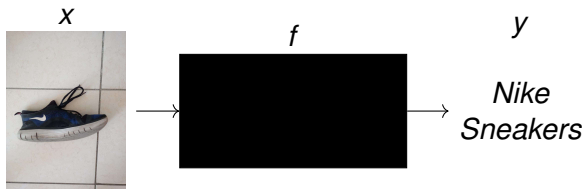
Subfields:

- Machine Learning (*ML*) & Deep Learning (*DL*).
- Computer Vision(*CV*).
- Natural Language Processing (*NLP*).
- Robotics.



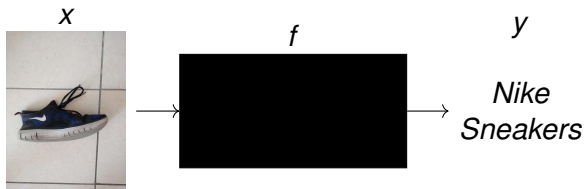
Explainable AI

We are interested in understanding models, behaving like a black box model:



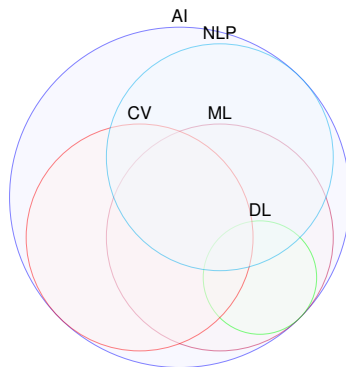
Explainable AI

We are interested in understanding models, behaving like a black box model:

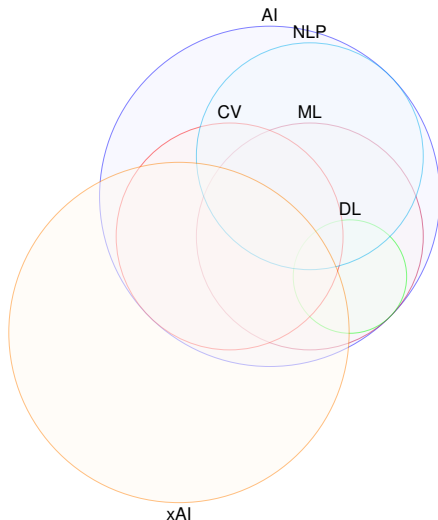


We want to *know why* $f(x) \rightarrow y$

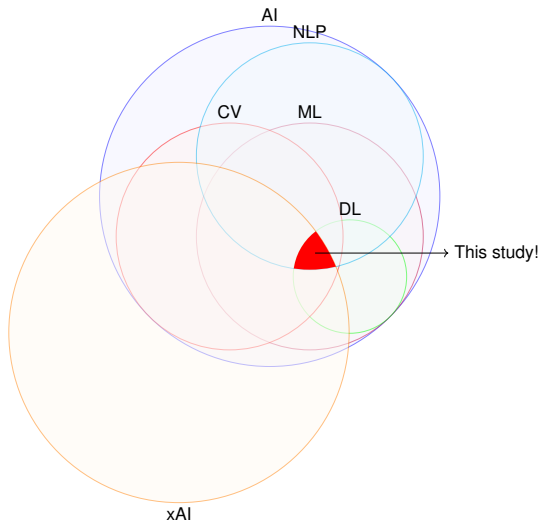
Fitting it all together



Fitting it all together



Fitting it all together



Thesis Objectives

Improvement of recognition and interpretable properties of model predictions.

Thesis Objectives

Improvement of recognition and interpretable properties of model predictions.

In particular:

- Development of low cost/complexity explainability approaches.
- Establishment of a fixed evaluation protocol.
- Differentiation of human based and machine explanations.

Table of Contents

- Introduction
- 1 Background**
- 2 Opti-CAM: Optimizing saliency maps for interpretability
- 3 CA-Stream: Attention-based pooling for interpretable image recognition
- 4 A learning paradigm for interpretable gradients
- References

Preliminaries

Desired data of Interpretability Study [4]

We ask questions
regarding *black box*
models.

- How does it work?
- How safe is it?
- Who is accountable for it?
- Who benefits from it?
- Who uses it?

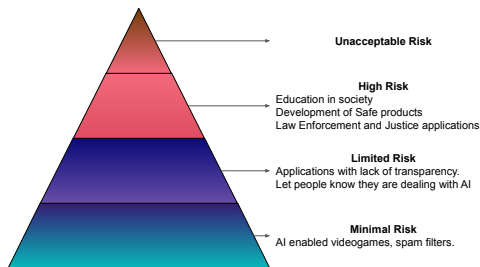
**Accepted AI systems
within society must
answer this.**

Preliminaries

Desired data of Interpretability Study [4]

We ask questions regarding *black box models*.

- How does it work?
- How safe is it?
- Who is accountable for it?
- Who benefits from it?
- Who uses it?



Regulation planned with the European AI act[3]

Accepted AI systems within society must answer this.

Image Recognition Models

Convolutional Neural Networks (CNN)

Based on the **convolution operation**.

A representation $f \star g$ is computed for a feature map f and a kernel g .
First approach with *Neocognitron* [5]

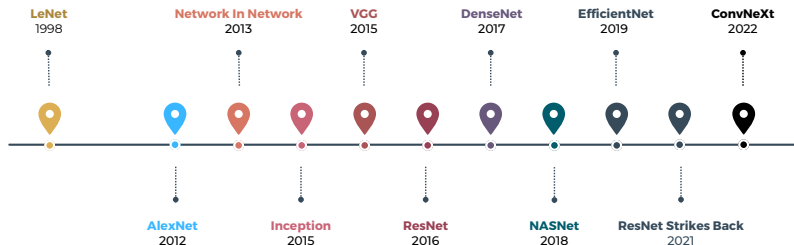
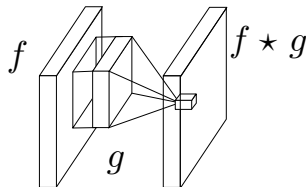


Image Recognition Models

Convolutional Neural Networks(CNN)

Training deep models is difficult, gradient information *is* lost with depth.

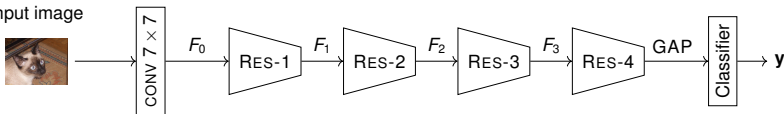
Image Recognition Models

Convolutional Neural Networks(CNN)

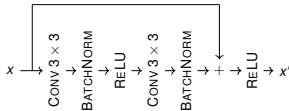
Training deep models is difficult, gradient information *is* lost with depth.

A model surviving the test of time: *ResNet* [6].

Input image



Basic Block



Bottleneck Block

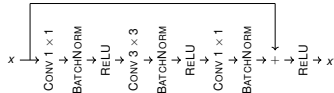


Image Recognition Models

Self Attention Architectures

Transformers

Attention: compute the relevance of each element relative to others in an embedding.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (1)$$

Image Recognition Models

Self Attention Architectures

An adoption of transformers, the *Vision Transformer* (ViT) [8]

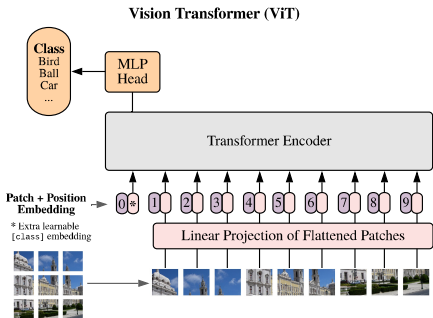
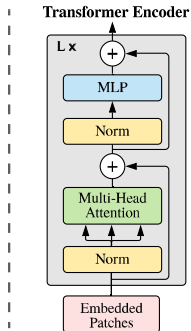
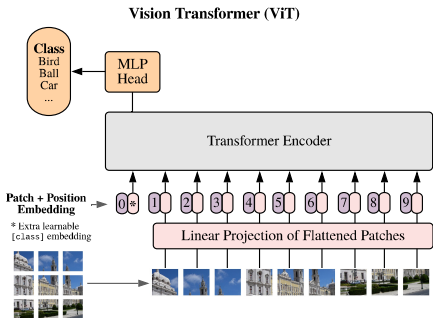


Image Recognition Models

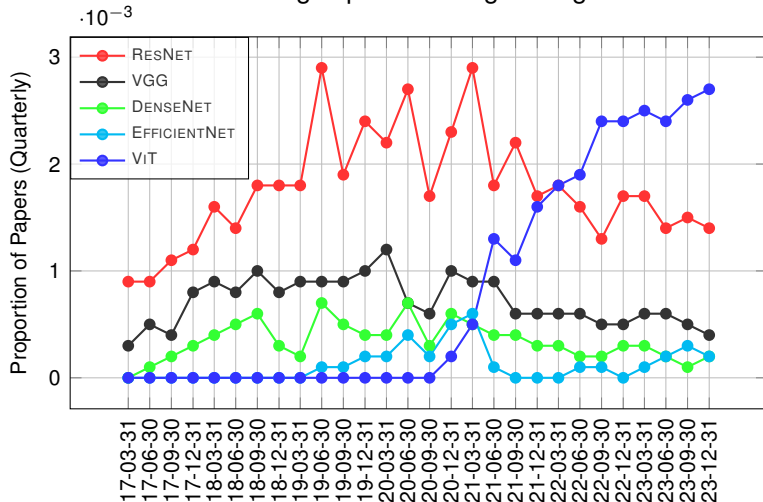
Self Attention Architectures

An adoption of transformers, the *Vision Transformer* (ViT) [8]



The Current Landscape

Transformers had a strong impact on image recognition:



Interpretability

Two dimension grouping

Interpretability

Two dimension grouping

Transparency

Approaches requiring changes to an architecture or training process.

Post-Hoc Interpretability

Approaches utilizing information contained within a model.

Interpretability

Class Activation Methods

Semantic information is found in deeper stages of a model [9].



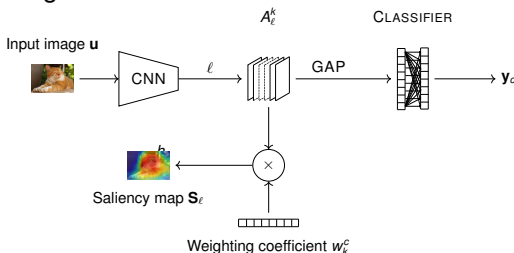
Interpretability

Class Activation Methods

Semantic information is found in deeper stages of a model [9].



Salient regions are highlighted for an input image:



Governed by equation:

$$S_\ell^c(\mathbf{u}) := h \left(\sum_k w_k^c A_\ell^k \right) \quad (2)$$

Interpretability

Evaluating Interpretability

Image Classification Approach

An explanation should demonstrate similar predictive properties to its query:

Input image. (I)



$$P_i^c = 0.8756$$

Explanation Map. (E^c)



$$O_i^c = 0.7442$$

Interpretability

Evaluating Interpretability

Image Classification Approach

An explanation should demonstrate similar predictive properties to its query:

Input image. (I)



$$P_i^c = 0.8756$$

Explanation Map. (E^c)



$$O_i^c = 0.7442$$

- Prediction loss when using an explanation. (Average Drop (AD) [10])
- Ratio of explanations being better than the input image. (Average Increase (AI) [10])
- Reaction to saliency guided modifications (Insertion/Deletion (Ins/Del) [11])

Interpretability

Evaluating Interpretability

Weakly Supervised Object Localization Approach

A saliency map should be good to localize the object of interest for an example:

Input image. (I)



Saliency Map. (E^c)



Interpretability

Evaluating Interpretability

Weakly Supervised Object Localization Approach

A saliency map should be good to localize the object of interest for an example:

Input image. (I)



Saliency Map. (E^c)

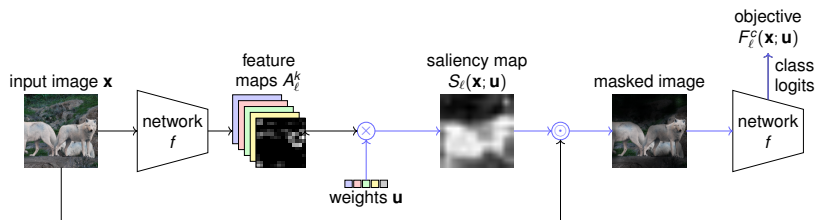


- Relationships between the bounding boxes (OM, LE, F_1).
- Overlap between object mask and bounding box (BA).
- Localization of most salient points and fraction of them in the bounding box (EP, SP).
- Penalty of the size of the bounding box and image size, regarding prediction (SM).

Preliminaries

Premise

Is it possible to optimize the weighting coefficient w_k^c to obtain a saliency map with the highest probability of prediction?



Opti-CAM

Motivation

● Combination of Feature Maps

Feature maps can be considered independent to compute w_k^c [12]. *What about a linear combination?*

$$F(\mathbf{w}) := f \left(\mathbf{u} \odot n \left(\text{up} \left(\sum_k w_k A_l^k \right) \right) \right)_c$$

Opti-CAM

Motivation

● Combination of Feature Maps

Feature maps can be considered independent to compute w_k^c [12]. *What about a linear combination?*

$$F(\mathbf{w}) := f \left(\mathbf{u} \odot n \left(\text{up} \left(\sum_k w_k A_l^k \right) \right) \right)_c$$

● Masking-based Saliency

Saliency maps produced by optimization in the input space [13, 14]

The optimization usually takes form of:

$$S^c(\mathbf{u}) := \underset{\mathbf{m} \in M}{\operatorname{argmax}} f(\mathbf{u} \odot n(\text{up}(\mathbf{m})))_c + \delta R(\mathbf{m})$$

Opti-CAM

Optimization

Formulation Assuming channel weights $w_k := \text{softmax}(\mathbf{u})_k$, the saliency map S_ℓ is a function of both an input image x and a variable $\mathbf{u}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

Opti-CAM

Optimization

Formulation Assuming channel weights $w_k := \text{softmax}(\mathbf{u})_k$, the saliency map S_ℓ is a function of both an input image x and a variable $\mathbf{u}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

Find vector $\mathbf{u}^*$ maximizing confidence for class c , with an input $\mathbf{x}$ masked by Saliency Map $S_\ell(\mathbf{x}; \mathbf{u}^*)$:

$$\mathbf{u}^* := \underset{\mathbf{u}}{\operatorname{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u}^*), \quad (4)$$

Opti-CAM

Optimization

Formulation Assuming channel weights $w_k := \text{softmax}(\mathbf{u})_k$, the saliency map S_ℓ is a function of both an input image x and a variable $\mathbf{u}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

Find vector $\mathbf{u}^*$ maximizing confidence for class c , with an input $\mathbf{x}$ masked by Saliency Map $S_\ell(\mathbf{x}; \mathbf{u}^*)$:

$$\mathbf{u}^* := \underset{\mathbf{u}}{\operatorname{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u}^*), \quad (4)$$

The objective function:

$$F_\ell^c(\mathbf{x}; \mathbf{u}) := g_c(f(x \odot S_\ell(\mathbf{x}; \mathbf{u}))) \quad (5)$$

Opti-CAM

Optimization

Formulation Assuming channel weights $w_k := \text{softmax}(\mathbf{u})_k$, the saliency map S_ℓ is a function of both an input image x and a variable $\mathbf{u}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

Find vector $\mathbf{u}^*$ maximizing confidence for class c , with an input $\mathbf{x}$ masked by Saliency Map $S_\ell(\mathbf{x}; \mathbf{u}^*)$:

$$\mathbf{u}^* := \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u}), \quad (4)$$

The objective function:

$$F_\ell^c(\mathbf{x}; \mathbf{u}) := g_c(f(x \odot S_\ell(\mathbf{x}; \mathbf{u}))) \quad (5)$$

Then:

$$S_\ell^c(\mathbf{x}) := S_\ell(\mathbf{x}; \mathbf{u}^*) = S_\ell(\mathbf{x}; \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u})) \quad (6)$$

Average Gain

Motivation

Existing metrics for measuring interpretability ($AD\downarrow$, $AI\uparrow$), can be fooled.

Average Gain

Motivation

Existing metrics for measuring interpretability ($AD\downarrow$, $AI\uparrow$), can be fooled.
All pixels are salient *but-one*:

Input



$$p_{389} = 0.874$$

Explanation



$$o_{389} = 0.822$$

- Drop = 5.95%
- Increase = 0%

Average Gain

Motivation

Existing metrics for measuring interpretability ($AD\downarrow$, $AI\uparrow$), can be fooled.
All pixels are salient *but-one*:

Input



$$p_{389} = 0.874$$

Explanation



$$o_{389} = 0.822$$

- Drop = 5.95%
- Increase = 0%

Average Gain

Definition

With the notation on slide (22):

$$\text{AG}(\%) := \frac{1}{N} \sum_{i=1}^N \frac{[o_i^c - p_i^c]_+}{1 - p_i^c} \cdot 100 \quad (7)$$

Measuring the increase in predictive power when masking an input.

Experimental Set-Up

Implementation

- All images are set to $224 \times 224 \times 3$.
- Adam [15] optimizer, for 100 iterations.
- Cross Entropy.
- Stop optimizing if loss is less than 10^{-10} .

¹Pytorch Model Zoo <https://pytorch.org/vision/0.8/models.html>

²Timm <https://github.com/rwightman/pytorch-image-models>

Experimental Set-Up

Implementation

- All images are set to $224 \times 224 \times 3$.
- Adam [15] optimizer, for 100 iterations.
- Cross Entropy.
- Stop optimizing if loss is less than 10^{-10} .

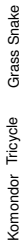
Datasets and Models

- **ILSVRC** Validation splits of ImageNet ([16, 17]), 50k images, over 1k classes.
- **Medical Data** Chest X-ray for 3 diseases [18]. Endoscopy data for pathologies, findings and landmarks [19].
- **Networks** ResNet-50 [6], and VGG [20]. ¹ViT-B (16-224) [8] and DeiT-B (16-224) [21] ².

¹Pytorch Model Zoo <https://pytorch.org/vision/0.8/models.html>

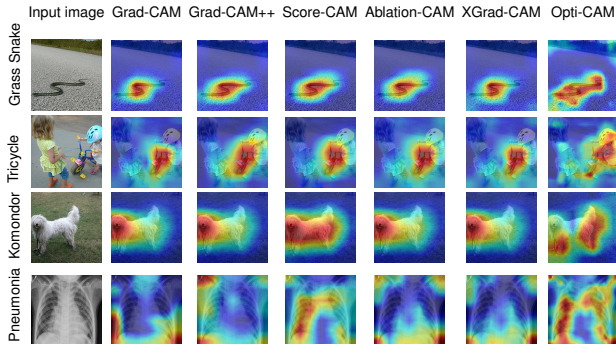
²Timm <https://github.com/rwightman/pytorch-image-models>

Qualitative Evaluation



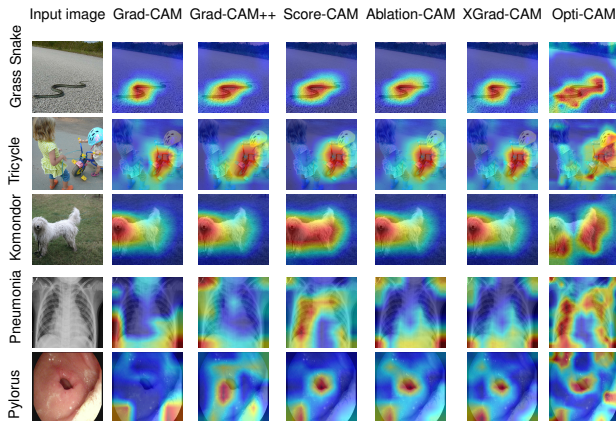
Results

Qualitative Evaluation



Results

Qualitative Evaluation



Results

Quantitative Evaluation

Classification Experiments:

METHOD	RESNET50						VGG16					
	AD ↓	AG ↑	AI ↑	I ↑	D ↓	T(s)	AD ↓	AG ↑	AI ↑	I ↑	D ↓	T(s)
Fake-CAM	0.8	1.6	46.0	50.7	28.1	0.00	0.5	0.6	42.6	46.1	26.9	0.00
Grad-CAM	12.2	17.6	44.4	66.3	14.7	0.03	14.2	14.7	40.6	64.1	11.6	0.02
Grad-CAM++	12.9	16.0	42.1	66.0	14.7	0.03	17.1	10.2	33.4	62.9	12.2	0.02
Score-CAM	8.6	26.6	56.7	65.7	16.3	15.22	13.5	15.6	41.7	62.5	12.1	3.11
Ablation-CAM	12.5	16.4	42.8	65.9	14.6	18.26	15.5	12.6	36.9	63.8	11.4	2.98
XGrad-CAM	12.2	17.6	44.4	66.3	14.7	0.03	13.8	14.8	41.2	64.1	11.7	0.02
Layer-CAM	15.6	15.0	38.8	67.0	14.2	0.08	48.9	3.1	13.5	58.3	6.4	0.07
ExPerturbation	38.1	9.5	22.5	70.7	15.0	152.96	43.0	7.1	20.5	61.1	15.0	83.20
Opti-CAM	1.5	68.8	92.8	62.0	19.7	4.15	1.3	71.2	92.7	59.2	11.0	3.94

Results

Quantitative Evaluation

Localization Experiments:

METHOD	RESNET50								VGG16							
	OM↓	LE↓	F1↑	BA↑	SP↑	EP↑	SM↓		OM↓	LE↓	F1↑	BA↑	SP↑	EP↑	SM↓	
Fake-CAM	63.6	54.0	57.7	47.9	99.8	28.5	0.98		64.7	54.0	57.7	47.9	99.8	28.5	1.07	
Grad-CAM	72.9	65.8	49.8	56.2	69.8	33.3	1.30		71.1	62.3	42.0	54.2	64.8	32.0	1.39	
Grad-CAM++	73.1	66.1	50.4	56.2	69.9	33.1	1.29		70.8	61.9	44.3	55.2	66.2	32.3	1.38	
Score-CAM	72.2	64.9	49.6	54.5	68.7	32.4	1.25		71.2	62.5	45.3	58.5	68.2	33.4	1.40	
Ablation-CAM	72.8	65.7	50.2	56.1	69.9	33.1	1.26		71.3	62.6	43.2	56.2	65.7	32.7	1.39	
XGrad-CAM	72.9	65.8	49.8	56.2	69.8	33.3	1.30		70.8	62.0	41.9	53.5	64.4	31.6	1.41	
Layer-CAM	73.1	66.0	50.1	55.5	70.0	33.0	1.29		70.5	61.5	28.0	54.7	65.0	32.4	1.45	
ExPerturbation	73.6	66.6	37.5	44.2	64.8	38.2	1.59		74.1	66.4	37.8	43.3	62.7	36.1	1.74	
Opti-CAM	72.2	64.8	47.3	49.2	59.4	30.5	1.34		69.1	59.9	44.1	51.2	61.4	30.7	1.34	

Table of Contents

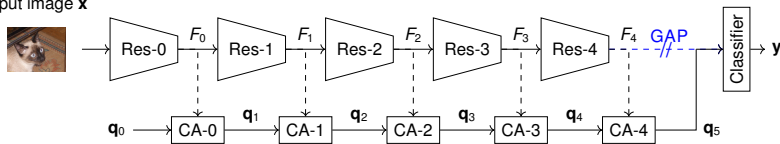
- Introduction
- 1 Background
- 2 Opti-CAM: Optimizing saliency maps for interpretability
- 3 CA-Stream: Attention-based pooling for interpretable image recognition**
- 4 A learning paradigm for interpretable gradients
- References

Preliminaries

Premise

The [CLS] token in transformers updates an abstract representation via attention with patch embeddings across the depth of a model. Is it possible to train a similar class representation on CNNs?

Input image x



Preliminaries

Contributions

- **CAM is Attention**

We demonstrate that CAM behaves in a similar manner as attention.

- **Cross Attention: [CLS] on CNNs**

We train an abstract class representation using a stream, alongside standard CNNs.

- **CAM on more architectures**

CAM is usually validated only on ResNet and VGG.
Extension to ConvNeXt.

CAM is Attention

Derivation

Let's freshen up:

Asuming $\mathbf{F}_\ell \in \mathbb{R}^{p_\ell \times d_\ell}$ a reshaping of $\mathbf{F}_\ell$, and $\mathbf{q}_\ell \in \mathbb{R}^{d_\ell}$ the CLS token³.

³ p relates to *width* $\times$ *height*. Spatial dimension

CAM is Attention

Derivation

Let's freshen up:

Asuming $\mathbf{F}_\ell \in \mathbb{R}^{p_\ell \times d_\ell}$ a reshaping of $\mathbf{F}_\ell$, and $\mathbf{q}_\ell \in \mathbb{R}^{d_\ell}$ the CLS token³.

Attention is defined as:

$$\mathbf{a} = \mathbf{A}^\top = \text{softmax} \left(\frac{\mathbf{F}_\ell \mathbf{q}_\ell}{\sqrt{d_\ell}} \right). \quad (8)$$

³ p relates to *width* $\times$ *height*. Spatial dimension

CAM is Attention

Derivation

Let's freshen up:

Asuming $\mathbf{F}_\ell \in \mathbb{R}^{p_\ell \times d_\ell}$ a reshaping of $\mathbf{F}_\ell$, and $\mathbf{q}_\ell \in \mathbb{R}^{d_\ell}$ the CLS token³.

Attention is defined as:

$$\mathbf{a} = \mathbf{A}^\top = \text{softmax} \left(\frac{\mathbf{F}_\ell \mathbf{q}_\ell}{\sqrt{d_\ell}} \right). \quad (8)$$

Upon replacing $\mathbf{q}_\ell$ with an arbitrary vector $\alpha \in \mathbb{R}^{d_\ell}$

$$\mathbf{a} = h_\ell(\mathbf{F}_\ell \alpha) = h_\ell \left(\sum_k \alpha_k \mathbf{f}_\ell^k \right). \quad (9)$$

With the activation function $h_\ell(\mathbf{x}) = \text{softmax}(\mathbf{x}/\sqrt{d_\ell})$

³ p relates to *width* $\times$ *height*. Spatial dimension

CAM is Attention

Pooling or Masking

We intend to integrate attention into a model, such that prediction and explanation are linked.

CAM is Attention

Pooling or Masking

We intend to integrate attention into a model, such that prediction and explanation are linked.

Cross Attention:

$$\text{CA}_\ell(\mathbf{q}_\ell, F_\ell) := F^\top \mathbf{a} = F_\ell^\top h_\ell(F_\ell \mathbf{q}_\ell) \in \mathbb{R}_\ell^d \quad (10)$$

CAM is Attention

Pooling or Masking

We intend to integrate attention into a model, such that prediction and explanation are linked.

Cross Attention:

$$\text{CA}_\ell(\mathbf{q}_\ell, F_\ell) := F^\top \mathbf{a} = F_\ell^\top h_\ell(F_\ell \mathbf{q}_\ell) \in \mathbb{R}_\ell^d \quad (10)$$

Which, reshaping the feature matrix as $F_\ell^\top = (\phi_\ell^1 \dots \phi_\ell^{p_\ell})$, a pooled representation can be expressed, via attention vector

$\mathbf{a} = (a_1, \dots, a_{p_\ell})$:

$$\text{CA}_\ell(\mathbf{q}_\ell, F_\ell) := F_\ell^\top \mathbf{a} = \sum_i a_i \phi_\ell^i. \quad (11)$$

CAM is Attention

Pooling or Masking

We intend to integrate attention into a model, such that prediction and explanation are linked.

Cross Attention:

$$\text{CA}_\ell(\mathbf{q}_\ell, F_\ell) := F^\top \mathbf{a} = F_\ell^\top h_\ell(F_\ell \mathbf{q}_\ell) \in \mathbb{R}_\ell^d \quad (10)$$

Which, reshaping the feature matrix as $F_\ell^\top = (\phi_\ell^1 \dots \phi_\ell^{p_\ell})$, a pooled representation can be expressed, via attention vector

$\mathbf{a} = (a_1, \dots, a_{p_\ell})$:

$$\text{CA}_\ell(\mathbf{q}_\ell, F_\ell) := F_\ell^\top \mathbf{a} = \sum_i a_i \phi_\ell^i. \quad (11)$$

A pooling similar to GAP, masked in a fashion similar to CAM.

Cross Attention Stream

Given a model f we consider locations where critical operations take place:

$$f = g \circ \text{GAP} \circ f_L \circ \dots f_0 \quad (12)$$

Cross Attention Stream

Given a model f we consider locations where critical operations take place:

$$f = g \circ \text{GAP} \circ f_L \circ \dots \circ f_0 \quad (12)$$

On another hand, A [CLS] token embedding is initialized, as $\mathbf{q}_0 \in \mathbb{R}^{d_0}$, updated with a sequence of embeddings $\mathbf{q}_\ell \in \mathbb{R}^{d_\ell}$ via interaction along a stream with features on stage ℓ :

$$\mathbf{q}_{\ell+1} = W_\ell \cdot \text{CA}_\ell(\mathbf{q}_\ell, F_\ell) \quad (13)$$

Cross Attention Stream

Given a model f we consider locations where critical operations take place:

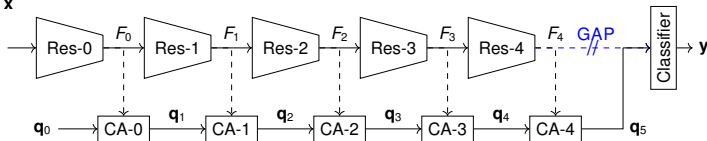
$$f = g \circ \text{GAP} \circ f_L \circ \dots \circ f_0 \quad (12)$$

On another hand, A [CLS] token embedding is initialized, as $\mathbf{q}_0 \in \mathbb{R}^{d_0}$, updated with a sequence of embeddings $\mathbf{q}_\ell \in \mathbb{R}^{d_\ell}$ via interaction along a stream with features on stage ℓ :

$$\mathbf{q}_{\ell+1} = W_\ell \cdot \text{CA}_\ell(\mathbf{q}_\ell, F_\ell) \quad (13)$$

Visualized as:

Input image $\mathbf{x}$



Experimental Set-Up

Training and models

- Training and validation in train and val sets of ILSVRC 2012 ([16, 17]) respectively.
- Using ResNet 50 original procedure [22]⁴.
- Training ResNet (18-50), ConvNeXt (Small, Base).
- Pascal-VOC 2012[23] and CUB[24] extensions.

⁴ <https://github.com/pytorch/vision/tree/main/references/classification>

Experimental Set-Up

Training and models

- Training and validation in train and val sets of ILSVRC 2012 ([16, 17]) respectively.
- Using ResNet 50 original procedure [22]⁴.
- Training ResNet (18-50), ConvNeXt (Small, Base).
- Pascal-VOC 2012[23] and CUB[24] extensions.

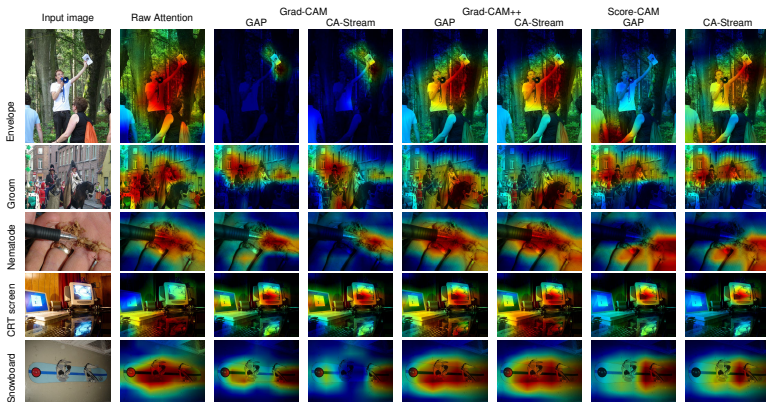
Evaluation

- Considering the whole image.
- No localization evaluation.
- Raw Attention used for OOD data.
- Ablation on parameters and design choices.

⁴ <https://github.com/pytorch/vision/tree/main/references/classification>

Results

Qualitative Experiments



Qualitative Experiments



Results

Quantitative Experiments

Recognition results

ACCURACY AND PARAMETERS					
NETWORK	POOL	GFLOPS	#PARAM	PARAM%	ACC↑
RESNET-18	GAP	3.648	11.69M	3.71	67.28
	CA	3.652	12.13M		67.54
RESNET-50	GAP	8.268	25.56M	27.27	74.55
	CA	8.288	32.53M		74.70
CONVNEXT-S	GAP	17.395	50.22M	1.95	83.26
	CA	17.400	51.20M		83.14
CONVNEXT-B	GAP	30.747	88.59M	1.96	83.72
	CA	30.753	90.33M		83.51

Results

Quantitative Experiments

Recognition: Small models

NETWORK	METHOD	POOL	AD↓	AG↑	AI↑	I↑	D↓
RESNET-18	Grad-CAM	GAP	17.64	12.73	41.21	63.13	10.66
		CA	16.99	17.22	44.95	65.94	10.68
	Grad-CAM++	GAP CA	19.05 19.02	11.16 14.76	37.99 40.82	62.80 65.53	10.75 10.82
CONVNEXT-S	Score-CAM	GAP	13.64	12.98	44.53	62.56	11.37
		CA	11.53	18.12	50.32	65.33	11.51
	Grad-CAM	GAP CA	42.99 22.09	1.69 14.91	12.60 32.65	48.42 84.82	30.12 43.02
	Grad-CAM++	GAP	56.42	1.32	10.35	48.28	33.41
		CA	51.87	9.40	20.55	84.28	52.58
	Score-CAM	GAP CA	74.79 64.21	1.29 8.81	10.10 18.96	47.40 82.92	38.21 57.46

Results

Quantitative Experiments

Recognition: Standard models

NETWORK	METHOD	POOL	AD↓	AG↑	AI↑	I↑	D↓
RESNET-50	Grad-CAM	GAP	13.04	17.56	44.47	72.57	13.24
		CA	12.54	22.67	48.56	75.53	13.50
	Grad-CAM++	GAP	13.79	15.87	42.08	72.32	13.33
		CA	13.99	19.29	44.60	75.21	13.78
	Score-CAM	GAP	8.83	17.97	48.46	71.99	14.31
		CA	7.09	23.65	54.20	74.91	14.68
CONVNEXT-B	Grad-CAM	GAP	33.72	2.43	15.25	52.85	29.57
		CA	19.45	13.96	32.89	86.38	45.29
	Grad-CAM++	GAP	34.01	2.37	15.60	52.83	29.17
		CA	36.69	8.00	21.95	85.39	53.42
	Score-CAM	GAP	43.55	2.23	15.67	50.96	39.49
		CA	23.51	11.04	27.35	83.41	60.53

Conclusions

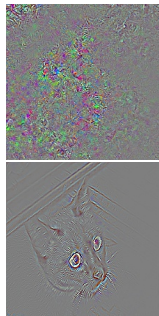
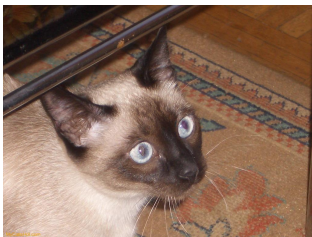
Table of Contents

- Introduction
- 1 Background
- 2 Opti-CAM: Optimizing saliency maps for interpretability
- 3 CA-Stream: Attention-based pooling for interpretable image recognition
- 4 A learning paradigm for interpretable gradients**
- References

Preliminaries

Premise

Gradient information is noisy and hard to interpret, however smoothing it makes it discriminative. Can gradient information be denoised during training, improving interpretability properties?



Preliminaries

Contributions

- Novel regularization for interpretable guided training.
- Improvement of post-hoc interpretations following transparency ideas.

Interpretable Gradients

Leveraging Backpropagation

The response (y) of a model (f) parameters (ϕ), after forwarding an input x can be visualized on the input space, via backpropagation:

$$\frac{\partial x}{\partial y} = \frac{\partial x}{\partial F_0} \frac{\partial F_0}{\partial F_1} \dots \frac{\partial F_{-1}}{\partial y} \quad (14)$$

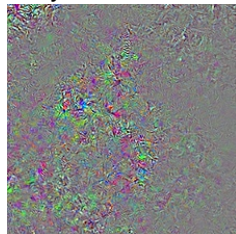
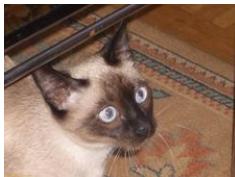
Interpretable Gradients

Leveraging Backpropagation

The response (y) of a model (f) parameters (ϕ), after forwarding an input x can be visualized on the input space, via backpropagation:

$$\frac{\partial x}{\partial y} = \frac{\partial x}{\partial F_0} \frac{\partial F_0}{\partial F_1} \dots \frac{\partial F_{-1}}{\partial y} \quad (14)$$

However, due to non-linearities it is noisy:



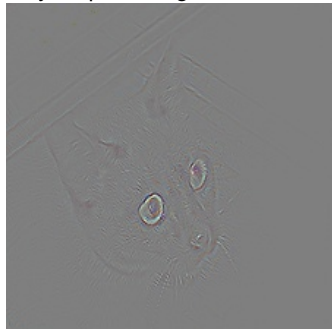
Interpretable Gradients

Soft Gradients

Soft gradients can be achieved by two approaches:

Guided Backpropagation [25]

Achieved by constraining backprop to only let positive gradients flow:



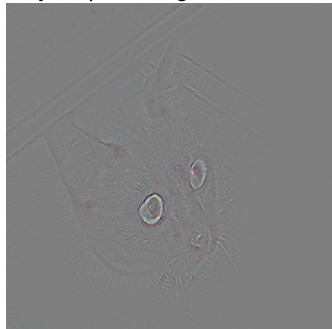
Interpretable Gradients

Soft Gradients

Soft gradients can be achieved by two approaches:

Guided Backpropagation [25]

Achieved by constraining backprop to only let positive gradients flow:



Smoothgrad [26]

Achieved by iteratively adding noise to the input space prior to forward pass:



Interpretable Gradients

Gradient as Regularization

Goal: Denoise the gradient during training using guided backprop.

How? Regularize using guided backprop.

Interpretable Gradients

Gradient as Regularization

Goal: Denoise the gradient during training using guided backprop.

How? Regularize using guided backprop.

Assuming target labels T :

$$L_R(X, \theta, T) = \frac{1}{n} \sum_{i=1}^n E(\delta x_i, \delta_G x_i), \quad (15)$$

Interpretable Gradients

Gradient as Regularization

Goal: Denoise the gradient during training using guided backprop.

How? Regularize using guided backprop.

Assuming target labels T :

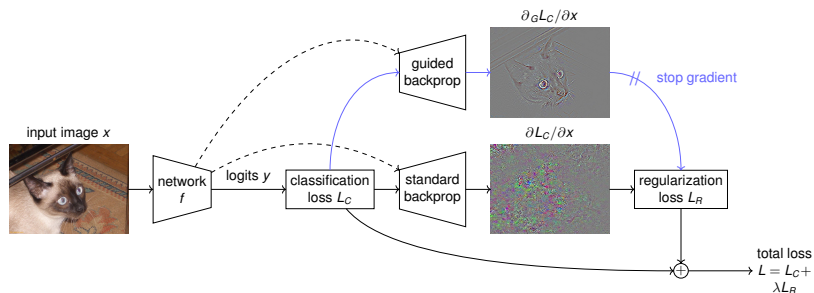
$$L_R(X, \theta, T) = \frac{1}{n} \sum_{i=1}^n E(\delta x_i, \delta_G x_i), \quad (15)$$

And the total loss:

$$L(x, \theta, t) = L_C(x, \theta, t) + \lambda L_R(x, \theta, t), \quad (16)$$

Interpretable Gradients

Formalized



Experimental Set-Up

Training and models

- Training and validation in train and val sets of CIFAR-100 ([27]) respectively.
- Using an optimized recipe for the dataset ⁵.
- Training ResNet (18), MobileNet [28].

⁵

Experimental Set-Up

Training and models

- Training and validation in train and val sets of CIFAR-100 ([27]) respectively.
- Using an optimized recipe for the dataset ⁵.
- Training ResNet (18), MobileNet [28].

Evaluation

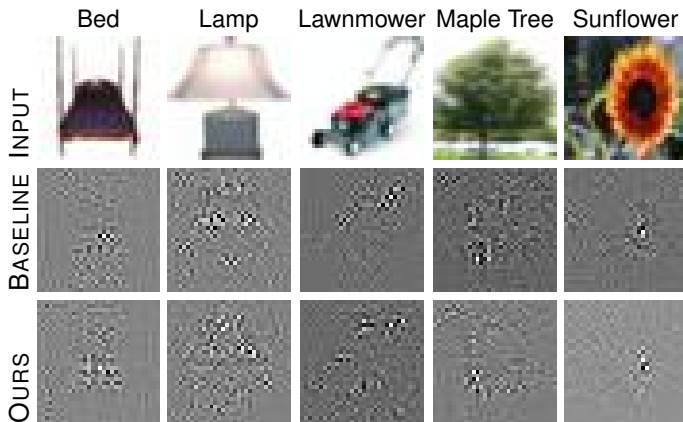
- CIFAR has no center crop. All image it is.
- No localization evaluation.
- Ablation on regularization functions.

⁵

Results

Qualitative Experiments

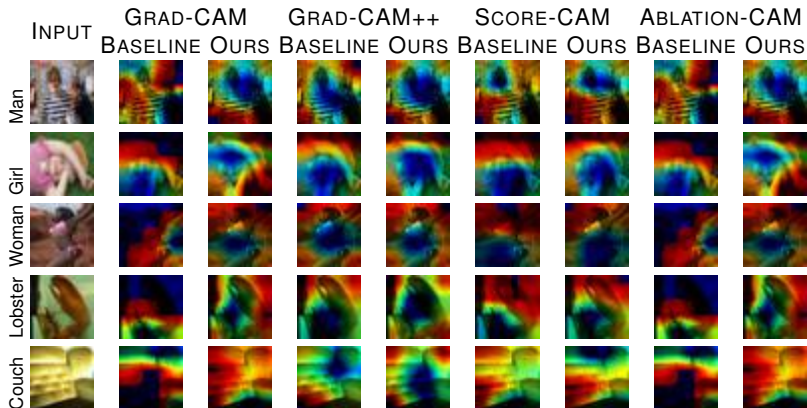
Denoising effect



Results

Qualitative Experiments

CAM Visualizations



Results

Quantitative Experiments

Recognition Metrics

RECOGNITION METRICS			
MODEL	ERROR	λ	Acc $\uparrow$
RESNET-18	-	-	73.42
	COSINE	7.5×10^{-3}	72.86
MOBILENET-V2	-	-	59.43
	COSINE	1×10^{-3}	62.36

Results

Quantitative Experiments

Interpretability metrics

Interpretable Recognition Metrics													
ResNet-18							MobileNet-V2						
Method	Error	AD↓	AG↑	AI↑	INS↑	DEL↓	Method	Error	AD↓	AG↑	AI↑	INS↑	DEL↓
GRAD-CAM	-	30.16	15.23	29.99	58.47	17.47	GRAD-CAM	-	44.64	6.57	25.62	44.64	14.34
	COSINE	28.09	16.19	31.53	58.76	17.57		COSINE	40.89	7.31	27.08	45.57	15.20
GRAD-CAM++	-	31.40	14.17	28.47	58.61	17.05	GRAD-CAM++	-	45.98	6.12	24.10	44.72	14.76
	COSINE	29.78	15.07	29.60	58.90	17.22		COSINE	40.76	6.85	26.46	45.51	14.92
SCORE-CAM	-	26.49	18.62	33.84	58.42	18.31	SCORE-CAM	-	40.55	7.85	28.57	45.62	14.52
	COSINE	24.82	19.49	35.51	59.11	18.34		COSINE	36.34	9.09	30.50	46.35	14.72
ABLATION-CAM	-	31.96	14.02	28.33	58.36	17.14	ABLATION-CAM	-	45.15	6.38	25.32	44.62	15.03
	COSINE	29.90	15.03	29.61	58.70	17.37		COSINE	41.13	7.03	26.10	45.38	15.12
AXIOM-CAM	-	30.16	15.23	29.98	58.47	17.47	AXIOM-CAM	-	44.65	6.57	25.62	44.64	15.27
	COSINE	28.09	16.20	31.53	58.76	17.57		COSINE	40.89	7.31	27.08	45.57	15.20

Conclusions

Table of Contents

- Introduction
- 1 Background
- 2 Opti-CAM: Optimizing saliency maps for interpretability
- 3 CA-Stream: Attention-based pooling for interpretable image recognition
- 4 A learning paradigm for interpretable gradients
- References

References I

-  J. Malik, P. Arbeláez, J. Carreira, K. Fragkiadaki, R. Girshick, G. Gkioxari, S. Gupta, B. Hariharan, A. Kar, and S. Tulsiani, “The three r’s of computer vision: Recognition, reconstruction and reorganization,” *Pattern Recognition Letters*, vol. 72, pp. 4–14, 2016.
-  J. McCarthy *et al.*, “What is artificial intelligence,” 2007.
-  T. Madiega, “Artificial intelligence act,” *European Parliament: European Parliamentary Research Service*, 2021.
-  Z. C. Lipton, “The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery.” *Queue*, vol. 16, no. 3, pp. 31–57, 2018.

References II

-  K. Fukushima, “Cognitron: A self-organizing multilayered neural network,” *Biological cybernetics*, vol. 20, no. 3-4, pp. 121–136, 1975.
-  K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
-  J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *arXiv preprint arXiv:1810.04805*, 2018.

References III

-  A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020.
-  H. Lee, R. Grosse, R. Ranganath, and A. Y. Ng, “Convolutional deep belief networks for scalable unsupervised learning of hierarchical representations,” in *Proceedings of the 26th annual international conference on machine learning*, 2009, pp. 609–616.
-  A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, “Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks,” in *WACV*, 2018.

References IV

-  P. Vitali, D. Abir, and S. Kate, “Rise: Randomized input sampling for explanation of black-box models,” *BMVC*, 2018.
-  H. Wang, M. Du, F. Yang, and Z. Zhang, “Score-cam: Improved visual explanations via score-weighted class activation mapping,” *CoRR*, vol. abs/1910.01279, 2019. [Online]. Available: <http://arxiv.org/abs/1910.01279>
-  R. C. Fong and A. Vedaldi, “Interpretable explanations of black boxes by meaningful perturbation,” in *ICCV*, 2017.
-  R. Fong, M. Patrick, and A. Vedaldi, “Understanding deep networks via extremal perturbations and smooth masks,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 2950–2958.

References V

-  D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *ICLR*, 2015.
-  A. Krizhevsky, I. Sutskever, and G. E. Hinton, “Imagenet classification with deep convolutional neural networks,” *Advances in neural information processing systems*, vol. 25, 2012.
-  O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, “ImageNet Large Scale Visual Recognition Challenge,” *International Journal of Computer Vision (IJCV)*, vol. 115, no. 3, pp. 211–252, 2015.
-  D. Kermany, K. Zhang, M. Goldbaum *et al.*, “Labeled optical coherence tomography (oct) and chest x-ray images for classification,” *Mendeley data*, vol. 2, no. 2, 2018.

References VI

-  K. Pogorelov, K. R. Randel, C. Griwodz, S. L. Eskeland, T. de Lange, D. Johansen, C. Spampinato, D.-T. Dang-Nguyen, M. Lux, P. T. Schmidt *et al.*, “Kvasir: A multi-class image dataset for computer aided gastrointestinal disease detection,” in *Multimedia Systems Conf.*, 2017.
-  K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” 2015.
-  H. Touvron, M. Cord, A. El-Nouby, P. Bojanowski, A. Joulin, G. Synnaeve, and H. Jégou, “Augmenting convolutional networks with attention-based aggregation,” *arXiv preprint arXiv:2112.13692*, 2021.

References VII

-  L. Bottou, “Large-scale machine learning with stochastic gradient descent,” in *Proceedings of COMPSTAT’2010: 19th International Conference on Computational Statistics Paris France, August 22-27, 2010 Keynote, Invited and Contributed Papers*. Springer, 2010, pp. 177–186.
-  M. Everingham, S. M. A. Eslami, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, “The pascal visual object classes challenge: A retrospective,” *International Journal of Computer Vision*, vol. 111, no. 1, pp. 98–136, Jan. 2015.
-  C. Wah, S. Branson, P. Welinder, P. Perona, and S. Belongie, “Caltech-ucsd birds-200-2011,” California Institute of Technology, Tech. Rep. CNS-TR-2011-001, 2011.

References VIII

-  J. T. Springenberg, A. Dosovitskiy, T. Brox, and M. Riedmiller, “Striving for simplicity: The all convolutional net,” *arXiv preprint arXiv:1412.6806*, 2014.
-  D. Smilkov, N. Thorat, B. Kim, F. Viégas, and M. Wattenberg, “Smoothgrad: removing noise by adding noise,” *arXiv preprint arXiv:1706.03825*, 2017.
-  A. Krizhevsky, “Learning multiple layers of features from tiny images,” pp. 32–33, 2009. [Online]. Available: <https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf>
-  A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, “Mobilenets: Efficient convolutional neural networks for mobile vision applications,” *CoRR*, vol. abs/1704.04861, 2017. [Online]. Available: <http://arxiv.org/abs/1704.04861>

References IX